Multi-Strategy Fusion Improved Northern Goshawk Optimizeris used for engineering problems and UAV Path Planning

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Abstract

Addressing the imbalance between exploration and exploitation, slow convergence, local optima Traps, and low convergence precision in the Northern Goshawk Optimizer (NGO): Introducing a Multi-Strategy Integrated Northern Goshawk Optimizer (MINGO).In response to challenges faced by the Northern Goshawk Optimizer (NGO), including issues like the imbalance between exploration and exploitation, slow convergence, susceptibility to local optima, and low convergence precision, this paper introduces an enhanced variant known as the Multi-Strategy Integrated Northern Goshawk Optimizer (MINGO). The algorithm tackles the balance between exploration and exploitation by improving exploration strategies and development approaches. It incorporates Levy flight strategies to preserve population diversity and enhance convergence precision. Additionally, to avoid getting trapped in local optima, the algorithm introduces Cauchy mutation strategies, improving its capability to escape local optima during the search process. Finally, individuals with poor fitness are eliminated using the crossover strategy of the Differential Evolution algorithm to enhance the overall population quality. To assess the performance of MINGO, this paper conducts an analysis from four perspectives: population diversity, balance between exploration and exploitation, convergence behavior, and various strategy variants. Furthermore, MINGO undergoes testing on the CEC-2017 and CEC-2022 benchmark problems. The test results, along with the Wilcoxon rank-sum test results, demonstrate that MINGO outperforms NGO and other advanced optimization algorithms in terms of overall performance. Finally, the applicability and superiority of MINGO are further validated on six real-world engineering problems and a 3D Trajectory planning for UAVs.

Key words: Northern Goshawk Optimizer, Exploration and exploitation, Heuristic algorithm, Engineering design problems, UAV Path Planning

1. Introduction

With the increasing diversity and complexity of engineering optimization problems in real-life scenarios, traditional optimization methods have proven insufficient in effectively addressing these challenges. In this context, metaheuristic optimization algorithms (MH) have experienced rapid development. Unlike traditional optimization methods, MH does not rely on gradient information of the solution space. Instead, it leverages random variables and natural heuristic rules to explore the search space in pursuit of optimal or near-optimal solutions[1]. As a result, metaheuristic algorithms exhibit strong capabilities in combating premature convergence and are relatively easy to implement for solving a wide range of optimization problems. They have found extensive applications across various domains, including computer science and medicine, attracting the attention of numerous researchers.

In addition to heuristic methods, numerical global optimization methods provide another approach to finding optimal solutions, supported by strict theoretical foundations for their convergence performance. Among them, evolutionary methods such as the Survival of the Fittest Algorithm (SoFA)are noteworthy global optimization techniques. SoFA is an optimization algorithm based on evolutionary strategies, simulating self-organizing features in biological evolution to optimize problems by continuously evolving new individuals[1]. Through interactions and adaptive mechanisms among individuals in the population, this algorithm gradually converges to the optimal solution during the search process. In addition to well-known heuristic algorithms, there are also strictly convergent stochastic optimization algorithms. Compared to heuristic methods, SoFA algorithm exhibits higher convergence and search efficiency, with its performance convergence verifiable through theoretical analysis[2].

In recent years, a plethora of bio-inspired optimization algorithms based on metaheuristic principles have been proposed. Depending on their sources of inspiration, metaheuristic algorithms can be categorized into four major classes[3]: These categories include Evolutionary Algorithms (EA), Physics-Based Algorithms (PhA), Swarm Intelligence-Based Algorithms (SI), and Human Behavior-Inspired Algorithms (HB)[4].

Evolutionary Algorithms (EA) belong to the first category and include Genetic Algorithms (GA), inspired by Charles Darwin's theory of evolution [5]; Differential Evolution (DE)[6]; and Evolution Strategies (ES)[7]. Among these, Genetic Algorithms (GA) and Differential Evolution (DE) are widely considered to be the most popular evolutionary algorithms, having garnered significant attention and being applied in numerous applications. For instance, they have been utilized in areas such as facial recognition[8], feature selection[9], network anomaly detection, data clustering[10], scheduling problems[11],and many other engineering applications and problem domains.

Physics-Based Algorithms (PhA) comprise the second category, and they include the Simulated Annealing algorithm (SA), inspired by the principles of solid-state annealing in metallurgy[12]; the Sine Cosine Algorithm (SCA)[13]; Multi-Verse Optimization algorithm (MVO)[14]; Galaxy Swarm Optimization (GSO) [15]; Firefly Algorithm (FFA) [16]; Firefly Algorithm (FFA) [17]; and Gravity Search Algorithm (GSA) [18].

The third category, Human Behavior-Inspired Algorithms (HB), encompasses metaheuristic algorithms designed to solve optimization problems by simulating certain natural behaviors of humans. Examples include the Teacher and Learner-Based Optimization Algorithm (TBLA)[19]; Social Evolution and Learning Optimization Algorithm (SELO)[20]and Soccer League Competition Algorithm (SLC)[21].

The fourth category, Swarm Intelligence-Based Algorithms (SI), includes metaheuristic algorithms that draw inspiration from the collective behavior of various animal species. Examples of SI algorithms include the Particle Swarm Optimization (PSO) algorithm, inspired by the foraging behavior of birds and fish [22]; Ant Colony Optimization (ACO) [23]; Grey Wolf Optimizer (GWO)[24]; Sea Star Algorithm (SSA) [25]; Whale Optimization Algorithm (WOA)[26];

Chameleon Swarm Algorithm (CSA)[27];Chimpanzee Optimization Algorithm (ChOA)[28]; Harris's Hawk Optimization algorithm (HHO)[29]; Marine Predator Algorithm (MPA) [30]; Artificial Gorilla Troop Optimization algorithm (GTO) [4]; Snake Optimizer (SO) [31];and the Northern Goshawk Optimizer (NGO)[32].These algorithms are among those proposed in the literature.Although these algorithms draw inspiration from different sources, they all encompass two crucial search phases: exploration and exploitation [33]. The exploration phase refers to the algorithm's exploration of various regions within the search space, striving to discover every promising area. The exploitation phase, on the other hand, entails continued searching within the potentially fruitful areas found during the exploration phase, seeking the optimal solution. A proficient MH (Metaheuristic) algorithm can strike a balance between these two phases, thereby achieving robust performance.

However, according to the No Free Lunch Theorem (NFL) [34], there exists no algorithm capable of solving all optimization problems. Each optimization algorithm is effective for some optimization problems while ineffective for others. Therefore, researchers continually propose various novel or improved optimization algorithms to address different optimization challenges. For instance, Kanovic et al. introduced the Generalized Particle Swarm Optimization (GPSO) algorithm [35] andZhang et al.[36]designed the Dynamic Grey Wolf Optimization Algorithm (DGWO). Liang et al. [37] proposed an enhanced Northern Goshawk Optimizer. Ma et al. [38], by imparting flight capabilities to grey wolves to enhance exploration, introduced an improved Grey Wolf Optimization Algorithm (AGWO). Additionally, Zhao et al. [39] put forward the Ocean Predator Algorithm based on quasi-oppositional learning and Q-learning (QQLMPA).

Optimization engineering problems pose significant challenges due to their diversity and complexity. These problems involve achieving objectives like minimizing costs or maximizing efficiency under limited resources [40]. They can be discrete or continuous, with multiple decision variables, constraints, and potential randomness or uncertainty. Selecting the right optimization algorithm is essential. While various algorithms, including traditional mathematical programming, heuristics, and evolutionary methods, have been developed, they still face issues such as slow convergence, limited precision, or getting trapped in local optima, particularly in complex scenarios [41].

The Northern Goshawk Optimizer (NGO), introduced by Mohammad Dehghani and colleagues, is inspired by the hunting behavior of the Northern Goshawk. It has a simple structure and performs well compared to other metaheuristic algorithms. However, NGO's search mechanism involves a prolonged random search, leading to high randomness and uncertainty. This can result in challenges such as an imbalance between exploration and exploitation, slow convergence, a tendency to get stuck in local optima, and low convergence accuracy in certain scenarios.

The paper addresses the shortcomings of the Northern Goshawk Optimization (NGO) algorithm and proposes a Multi-Strategy Integrated Northern Goshawk Optimization (MINGO) algorithm. The main contributions of this paper are as follows:

(1) Levy Flight Strategy: By introducing the Levy flight exploration strategy, it enhances the algorithm's ability to maintain population diversity, prevents it from getting trapped in local optima, and improves convergence speed and accuracy.

(2) Brownian Motion Strategy: Adding randomness helps the algorithm explore the solution space more effectively and increases the likelihood of escaping local optima.

(3) Reinforced exploration Strategy: Improving the position update formulas during both exploration and exploitation processes to balance the algorithm's local exploitation performance and global search capability.

(4) Cauchy Mutation Strategy: Enhancing the efficiency, stability, and global search capability of the optimization algorithm, aiding in finding the optimal solution or better solutions to problems more quickly.

(5) To evaluate MINGO's performance, we conducted comparative tests with 12 popular algorithms on the CEC-2017 and CEC-2022 benchmark test functions.

(6) We validated MINGO's effectiveness and accuracy in solving real-world engineering design problems through six engineering design cases anda 3D Trajectory planning for UAVs.

This paper is organized as follows: Section 2 gives a brief introduction to the NGO algorithm. Section 3 details the improvement strategy of MINGO and systematically describes the MINGO algorithm. In Section 4, we conduct corresponding simulation experiments and detailed analysis. Section 5 We apply the proposed MINGO to six classical practical engineering design problems and a 3D Trajectory planning for UAVs. Section 6 concludes the paper with a summary and outlook.

2. Improved Northern Goshawk Optimizer

This section provides a detailed description of the four enhancement strategies we propose, which include: the Levy flight strategy, the Brownian motion strategy, the Cauchy mutation strategy, and the augmented exploration strategy. Additionally, we have conducted a time complexity analysis.

2.1.NGO inspiration and motivation for improvement

The NGO algorithm exhibits better optimization capabilities compared to many previously proposed metaheuristic algorithms. However, its high degree of randomness during the search process can lead to an imbalance between global exploration and local exploitation. An analysis of NGO's exploration and exploitation reveals that it primarily focuses on global exploration for the majority of test functions [32]. However, in the later stages of the search process, it should shift towards deep exploration within local regions. Without local exploration, the search may get trapped in local optima, affecting the accuracy of the optimal solution. Furthermore, although NGO maintains a high population diversity

throughout, a high population diversity doesn't necessarily guarantee good search results. To address these limitations, a Multi-Strategy Integrated NGO (MINGO) algorithm is proposed.

2.2. Levy flight strategy

To tackle issues like insufficient population diversity, high randomness, and premature convergence in the Northern Goshawk Optimizer (NGO), the Lévy flight strategy[41] is introduced.This random walk model, with step lengths following a heavy-tailed stable distribution, enhances randomness and helps the algorithm escape local optima. Larger step lengths expand the search range, improving population diversity, while smaller step lengths enhance optimization accuracy.

Leveraging the Lévy flight strategy, we apply it to the exploration stage of the search process. In competitive environments, individuals with better adaptability and survivability excel, similar to Northern goshawks, which compete for resources to ensure survival. Based on this, we propose a new formula to update candidate solution byEq.(1).

$$x_{i,j}^{new,P1} = x_{best} + RL \times (x_{best_{i,j}} - I \times x_{i,j}) \tag{1}$$

$$RL = 0.5 \times Levy(Dim) \tag{2}$$

In the equation, $x_{best_{i,j}}$ represents the best solution obtained up to the t^{th}iteration, RLdenotes the Levy flight with assigned weights, Dim represents the dimension space, and Levy(Dim)is the Levy flight distribution function. The calculation method is illustrated in Eq. (3).

$$Levy(D) = s \times \frac{u \times \sigma}{|v|^{\frac{1}{\eta}}} \tag{3}$$

where sis a constant fixed at 0.01，η is a constant fixed at 1.5. uand vare random numbers within the interval [0,1].The calculation of σis determined by Eq.(4).

$$\sigma = \left(\frac{\Gamma(1+\eta) \times sin\left(\frac{\pi\eta}{2}\right)}{\Gamma\left(\frac{1+\eta}{2}\right) \times \eta \times 2\left(\frac{\eta-1}{2}\right)} \right)^{\frac{1}{\eta}} \tag{4}$$

where, Γ represents the gamma function. The value of ηremains the same, set at 1.5.

2.3. Brownian motion strategy

To address the issues of Northern Goshawk Optimizer easily getting trapped in local optima, struggling with highly nonlinear, non-convex, or multimodal problems, we introduce the Brownian motion strategy in the global search phase [42]To address the issues of Northern Goshawk Optimizer easily getting trapped in local optima, struggling with highly nonlinear, non-convex, or multimodal problems, we introduce the Brownian motion strategy in the global search phase in the exploration stage. The updated position update formula is presented in Eq.(5).

$$x_{i,j}^{new,P1} = x_{i,j} + RB \times \left(x_{i,j} - p_{i,j}\right) \tag{5}$$

where, RB is a random number that conforms to a Brownian distribution.

Taking into account the improvements described in Sections 2.2 and 2.3, the updated position update formula for the first stage of Northern goshawk hunting can be represented by Eq.(6).

$$x_{i,j}^{new,P1} = \begin{cases} x_{best} + RL \times (x_{best} - I \times x_{i,j}), & F_{P_i} < F_i \\ x_{i,j} + RB \times \left(x_{i,j} - p_{i,j}\right), & F_{P_i} \geq F_i \end{cases} \tag{6}$$

2.4. Cauchy mutation strategy

The Northern Goshawk Optimizer tends to get trapped in local optima in the later stages of optimization. To address this issue, we introduce the Cauchy mutation operator. The Cauchy mutation is derived from the Cauchy distribution [43], which is a common continuous probability distribution in probability theory. It has a high probability density in the middle and low probability density at the ends. Due to the characteristic of having long and flat tails on both sides, the Cauchy mutation possesses strong perturbation capabilities. The probability density formula for the standard Cauchy distribution is shown in Eq.(7).

$$f(x) = \frac{1}{\pi(x^2+1)} x \in (-\infty, +\infty) \tag{7}$$

The Cauchy distribution is characterized by its peak at the origin and a gradual extension towards both ends[44]. The introduction of Cauchy mutation in the Northern Goshawk Optimizer involves perturbing and mutating the best solution under the current circumstances to obtain a new solution, thereby reducing the risk of the original algorithm getting trapped in a local optimum. "Cauchy" represents the Cauchy mutation operator, and its calculation method is shown inEq.(8). Therefore, the updated position update formula is given by Eq.(9).

$$Cauchy = 1 + tan\left(0.5 \times \pi \times (rand - 0.5)\right) \tag{8}$$

$$x_{i,j}^{new,P2} = x_{i,j} + (-R + 2 \times RB) \times x_{i,j} \times Cauchy \tag{9}$$

2.5.Enhanced exploitation strategy

In this section, we introduce a new exploitation strategy that enhances the original NGO's exploitation capability,

accelerates algorithm convergence, and improves convergence accuracy, making it more effective in handling high-dimensional complex problems. The new exploitation formula is represented as Eq.(10).

$$x_{i,j}^{new,P2} = x_{i,j} + RB \times (p_{i,j} - I \times x_{i,j}) \tag{10}$$

where, we have set a balance factor($R_1 = 0.5$). When$rand < R_1$,Eq.(9)is employed to explore the search space; otherwise, our newly proposed Eq. (10) is used.

Combining equations 3.4 and 3.5, the improved position update formula for the second stage of Northern goshawk hunting can be represented as Eq.(11).

$$x_{i,j}^{new,P2} = \begin{cases} x_{i,j} + (-R + 2 \times RB) \times x_{i,j} \times cauchy, & rand < R_1 \\ x_{i,j} + RB \times (p_{i,j} - I \times x_{i,j}), & else \end{cases} \tag{11}$$

In summary, the pseudocode for MINGO is presented in Table. 1.

Table. 1. Pseudocode for MINGO.

Algorithm 1 Pseudo Code of MINGO

```
Initialize Problem Setting (Dim, ub,lbPop_size(N)), Max_Iter(T), Curr_Iter(t)
Initialize the population randomly
For t = 1:T
Calculate the fitness of the search agents.
Step 1: prey identification (exploration)
Random hunting for prey
    if fit(i) > F_P
        Update the positions of northern goshawk using Eq. (1)
else
        Update the positions of northern goshawk using Eq. (5)
    end if
    Update i^th positions of the northern goshawk
  Step 2: chase prey (exploitation)
        Calculate Cauchy using Eq.(8)
    if rand < 0.5
            Update the positions of northern goshawk using Eq. (9)
        else
  Update the positions of northern goshawk using Eq. (10)
    end if
    Update i^th positions of the northern goshawk
          t = t + 1
    end for  t = 1:T
Return the best solution
end
```

2.6.Time Complexity

Different algorithms targeting the same optimization problem may exhibit varying execution times. An important way to evaluate algorithm execution time is to consider their computational complexity. In this article, we employ the Big O notation [45] to analyze and compare the time complexities of NGO and MINGO. During the initialization phase, the fitness evaluation time for each Northern Sparrowhawk is$O(N)$, whereas during the iteration process, each Northern Sparrowhawk needs to undergo two phases of updates. Hence, the computational complexity for these two phases is $O(2 \times N \times T \times Dim)$. Here, Nrepresents the population size of Northern Sparrowhawks, T denotes the maximum number of iterations, and Dimsignifies the dimensionality of the search space. Therefore, the total computational time for the NGO algorithm is $O(N \times (2 \times T \times Dim + 1))$[32]. In the case of MINGO, we do not introduce any additional loop iterations, and as a result, its complexity remains $O(N \times (2 \times T \times Dim + 1))$.

3. Experimental results and analysis

In this section, we have designed three different experiments to evaluate the performance of the proposed MINGO algorithm. Firstly, we assess the algorithm's effectiveness by measuring its exploration and exploitation capabilities, population diversity, and variations of the proposed strategies. Secondly, we conduct comparative tests of this algorithm against 12 other algorithms in both CEC-2017 and CEC-2022 to validate its performance. Lastly, we subject it to rank-sum tests and Friedman tests to ascertain whether there are significant differences in performance between the MINGO algorithm and other algorithms. All experiments were conducted on a desktop computer equipped with a 13th Intel(R) Core (TM) i5-13400 (16 CPUs), ~2.5 GHz, and 16 GB RAM, using the MATLAB 2022b platform.

3.1. MINGO Qualitative analysis

In this section, the performance of MINGO is validated using the CEC-2017 test suite in a 30-dimensional space, focusing on four aspects: exploration and exploitation, convergence behavior, population diversity, and variations of different strategies. The CEC-2017 test suite functions are listed in Appendix 1for reference.

3.1.1. Analysis of MINGO exploration and exploitation performance

In metaheuristic algorithms, exploration and exploitation are crucial. Exploration seeks new solutions in uncharted areas, while exploitation focuses on improving known solutions within local neighborhoods. A balanced approach enhances the algorithm's efficiency, adaptability, and robustness across various optimization problems [46]. An excellent algorithm should strike a good balance between these two factors. Therefore, we calculate the percentages of exploration and exploitation using Eq.(12)and Eq.(13) respectively. $Div(t)$ is a measure of dimensional diversity, computed usingEq.(14).Here, x_{id} represents the position of agent i^{th} in dimension d, and Div_{max} represents the maximum diversity throughout the entire iteration process.

$$Exploration(\%) = \frac{Div(t)}{Div_{max}} \times 100 \tag{12}$$

$$Exploitation(\%) = \frac{|Div(t) - Div_{max}|}{Div_{max}} \times 100 \tag{13}$$

$$Div(t) = \frac{1}{D}\sum_{d=1}^{D}\frac{1}{N}\sum_{i=1}^{N}\left|median\left(x_d(t)\right) - x_{id}(t)\right| \tag{14}$$

Fig. 1illustrates the balance between exploration and exploitation in MINGO on the 30-dimensional CEC-2017 test functions. The MINGO algorithm initially focuses on global exploration and then shifts to local exploitation, achieving a balance by the mid-stage of iterations. Notably, it maintains a high exploitation ratio throughout, enhancing convergence speed and search precision. This dynamic equilibrium helps MINGO avoid local optima and premature convergence.

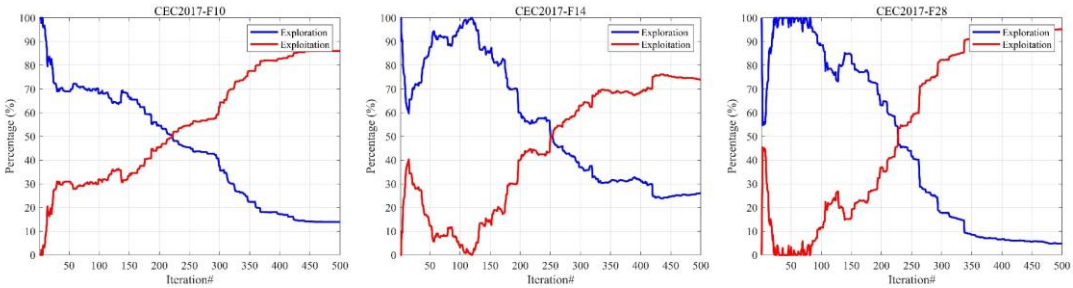

Fig. 1. Balance between exploration and exploitation.

3.1.2. Convergence behavior analysis

To validate the convergence characteristics of MINGO, we designed experiments to analyze its convergence behavior. As shown in Fig. 2.The first column shows the two-dimensional search space of the test functions, highlighting its complexity. The second column depicts the search trajectories of agents, who cluster around the optimal solution and cover the entire space, demonstrating MINGO's ability to avoid local optima. The third column illustrates the variation in average fitness values, which initially high, rapidly decreases as agents approach the optimal solution. The fourth column shows agents' trajectories shifting from global exploration to local exploitation. The final column displays MINGO's convergence curves: for unimodal functions, the curve steadily decreases, while for multimodal functions, it exhibits a step-like descent, indicating MINGO's capability to escape local optima and find the global optimum.

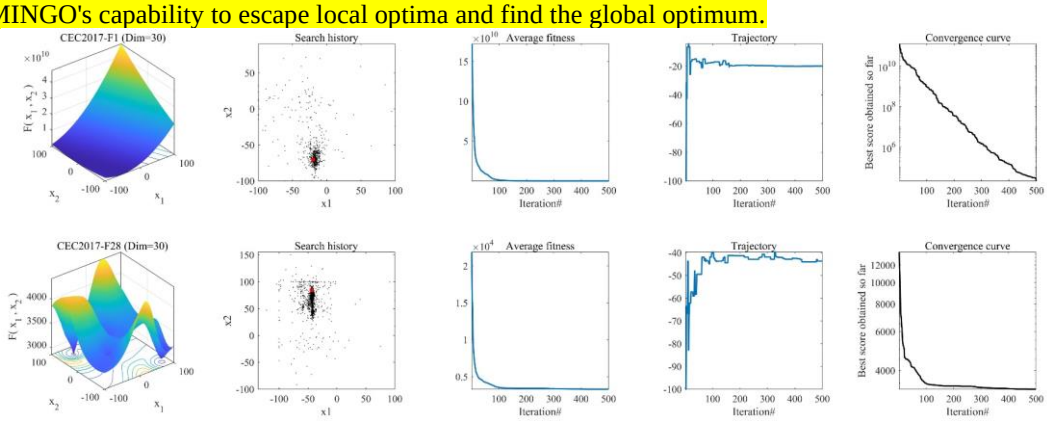

Fig. 2. MINGO convergence behavior.

3.1.3. Population diversity analysis

Population diversity plays a crucial role in metaheuristic algorithms. A highly diverse population indicates significant differences among individuals within the population, enabling a broader exploration of the search space and preventing premature convergence to local optima. Thus, maintaining population diversity is a fundamental objective in metaheuristic algorithms [47].Typically, the following formula is employed to measure the population diversity of an algorithm:

$$I_c(t) = \sqrt{\sum_{i=1}^{N}\sum_{d=1}^{D}(x_{id}(t) - c_d(t))^2} \tag{15}$$

$$c_d(t) = \frac{1}{D} \times \sum_{i=1}^{N} x_{id}(t) \tag{16}$$

The calculation method described by Morrison in 2002[48]measures the dispersion of individuals from the population's centroid, where$I_c$represents this dispersion,$c_d$is the centroid, and$x_{id}$is the value of thed^{th}dimension of thei^{th}individual. While diverse populations can reduce early convergence risks and local optima, excessive diversity may hinder convergence speed and accuracy.Fig. 3shows MINGO's population diversity on CEC-2017 benchmark functions across various dimensions. The diversity increases in later iterations due to the Lévy flight and Cauchy mutation strategies, which enhance diversity during the early to mid-stages of the search.

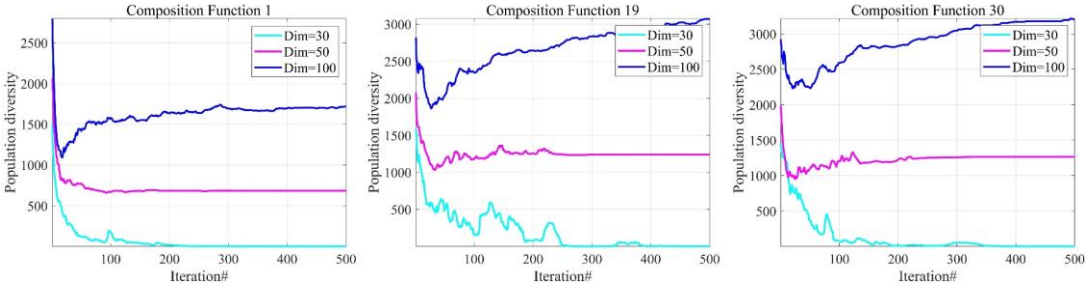

Fig. 3. Population diversity of MINGO in different dimensions in CEC-2017.

3.1.4. The effect of different strategies

To enhance the performance of NGO, we proposed four strategies. In this section, we will validate the effectiveness of these four strategies. During the improvement process, we sequentially addressed the algorithm's shortcomings. Therefore, we will construct algorithm variants by combining these strategies one by one. We compared these variants based on the 30-dimensional CEC-2017 benchmark functions. The compared algorithms include:NGO (No Improvement), LNGO (Levy Flight Strategy Only), BNGO (Levy Flight Strategy and Brownian Motion Strategy), CNGO (Levy Flight Strategy, Brownian Motion Strategy, and Cauchy Mutation Strategy) and MINGO (All Four Strategies Combined). To ensure the validity of the experiments, all algorithms were independently run 30 times. The parameter settings for the variant algorithms were consistent with those of NGO.

The experimental results are shown in Appendix 2. From the table, it can be observed that MINGO, which combines all three strategies, exhibits the best overall performance with 21 optimal mean values achieved. Based on the data in the table, we can observe that in general, the performance of NGO, LNGO, BNGO, CNGO, and MINGO increases sequentially.However, for LNGO, BNGO, and CNGO, their performance on certain functions is not as good as NGO. This is because the improved search strategies were initially designed to strike a balance between exploration and exploitation, and to some extent, they enhance convergence speed. While the standalone improvement in exploration strategy may not perform well, the combination of all four strategies proves to be effective. It's important to note that algorithm performance can vary depending on the specific problem or function being optimized. The choice of strategy or combination of strategies may need to be tailored to the characteristics of the optimization problem at hand. Fig. 4 presents the convergence curves of the three strategy variant algorithms on selected functions. From the graph, it becomes clearer that the algorithm's performance gradually improves as more strategies are incorporated. Moreover, it's evident from the graph that during the exploration phase, LNGO, BNGO, and CNGO exhibit faster convergence compared to NGO.Although MINGO didn't achieve ideal results on certain functions, it's important to note that the No Free Lunch theorem for optimization (NFL theorem) states that there is no universal algorithm that can solve all optimization problems [34]. In summary, combining and applying the four strategies to improve NGO is effective. It not only enhances the algorithm's convergence speed but also improves its convergence accuracy.

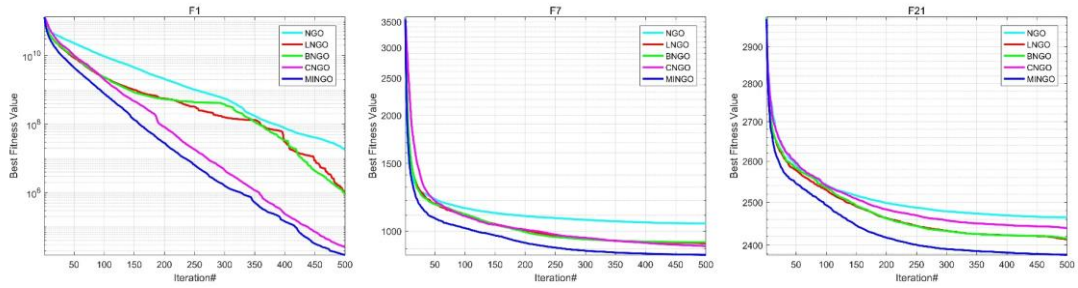

Fig. 4.Convergence curves of algorithms with different strategy variants.

3.2. MINGO quantitative analysis

In this section, we compare MINGO with other 12 algorithms in CEC-2017[49] and CEC-2022[50] to verify its performance.

3.2.1. Competing algorithms and parameter Settings

In this section, to validate the effectiveness of MINGO, we will compare it with 12 state-of-the-art algorithms on both CEC-2017 and CEC-2022 benchmark functions. The compared algorithms includeSaDE[51],AVOA[52], DBO[53], GSA[18], GWO[24], AO[3], MFO[54], WOA[26], HHO[29], SO[31], CPSOGSA[55], CSA[27]and NGO[32]. We will use CEC-2017 benchmark functions for the comparison. The parameter settings for the compared algorithms are provided in Table. 2. All algorithms will be independently run 30 times, and the experimental results will be presented in the following sections. The best results for each test function and its corresponding dimension will be highlighted in bold.

Table. 2. Compare the parameter Settings of the algorithms.

Algorithms	Name of the parameter	Value of the parameter
SaDE	p1, CRm, ns1, ns2	0.5, 0.5, 0, 0
AVOA	L1, L2, w, p1, p2, p3	0.8, 0.2, 2.5, 0.6, 0.4, 0.6
DBO	P_percent	0.2
GSA	ElitistCheck , min_flag ,Rpower , Rnorm	1 ,1 ,1 ,2
GWO	a	[0,2]
AO	alpha, delta	0.1， 0.1
MFO	b, t	1, [-1,1]
WOA	a, a2, b	[0,2], [-1,-2], 1
HHO	E0, E1	[-1,1], [0,2]
SO	Q, T, c1, c2, c3	0.25, 0.6, 0.5, 0.05, 2
CPSOGSA	φ1, φ2	2.05, 2.05
CSA	γ、α、β	1, 3.5, 3
NGO	R, I	Linearly decreased from 0.02 to 0, 1 or 2

3.2.2. CEC-2017 experimental results

In this section, for a better evaluation of MINGO's performance, we conducted comparative validations using the CEC-2017 benchmark functions, as illustrated in Appendix 1. The CEC-2017 benchmark functions consist of four categories: single-peaked functions, multi-peaked functions, hybrid functions, and composition functions. Among these, single-peaked functions possess only one global optimum, without any local optima, making them suitable for assessing the development performance of the algorithm. Multi-peaked test functions, on the other hand, feature multiple local optima, primarily serving to evaluate the algorithm's ability to discover the global optimum and escape local optima. The latter two categories of functions are used to measure the algorithm's capability in handling complex continuous problems.

When validating on the CEC-2017 test suite, we conducted tests in dimensions of 30, 50, and 100. The results for different dimensions are presented in Appendix3toAppendix 5. Convergence curves for selected functions are displayed in Fig. 5, Fig. 6 and Fig. 7.

Based on the experimental results, it is evident that the MINGO algorithm consistently maintains a leading position across 30, 50, and 100 dimensions. In the case of the 30-dimensional tests, MINGO achieves the best average ranking for 18 functions, the second-best average ranking for 5 functions, and the third-best average ranking for 4 functions, with no instances of being the worst performer. As the dimensionality increases, MINGO continues to deliver strong optimization results. In the 50-dimensional and 100-dimensional scenarios, MINGO achieves the best average ranking for 21 functions and the second-best average ranking for 3 functions, without any instances of being the worst performer.

Fig. 8 displays the ranking of all algorithms across different dimensions. To enhance visualization, we categorized rankings into five major groups: Best Average Ranking, Second Best Average Ranking, Third Best Average Ranking, Worst Average Ranking, and Other Rankings. From the graph, it is evident that MINGO consistently avoids the Worst Average Ranking category across all three dimensions, and its overall rankings do not differ significantly from other algorithms.

From the convergence plots in different dimensions, it is evident that algorithms such as SO, WOA, AO, HHO, and NGO tend to get stuck in local optima during the later iterations, lacking the ability to escape from local optima. In contrast, MINGO maintains a strong exploration capability even in the later stages of iterations. While functions F1, F2, and F13 temporarily converge to local optima during a certain period of iterations, MINGO manages to escape from these local optima in the later iterations and continues to explore deeply, ultimately achieving higher convergence accuracy. This indicates the effectiveness of the Brownian motion strategy and Levy flight strategy that we introduced. These strategies not only help the algorithm escape from local optima but also enhance the algorithm's convergence speed and precision.

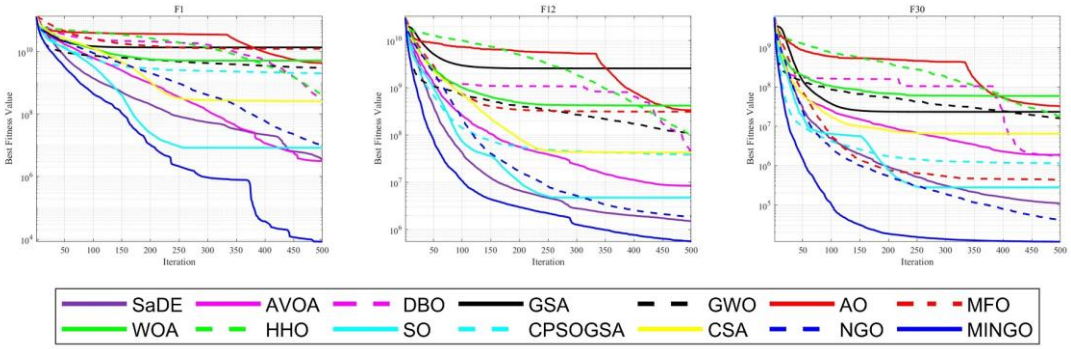

Fig. 5. CEC-2017 test function comparison curve (Dim=30).

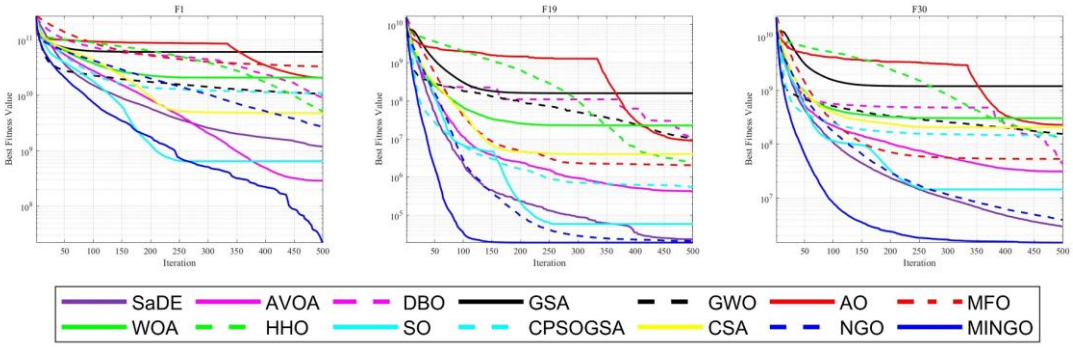

Fig. 6. CEC-2017 test function comparison curve (Dim=50).

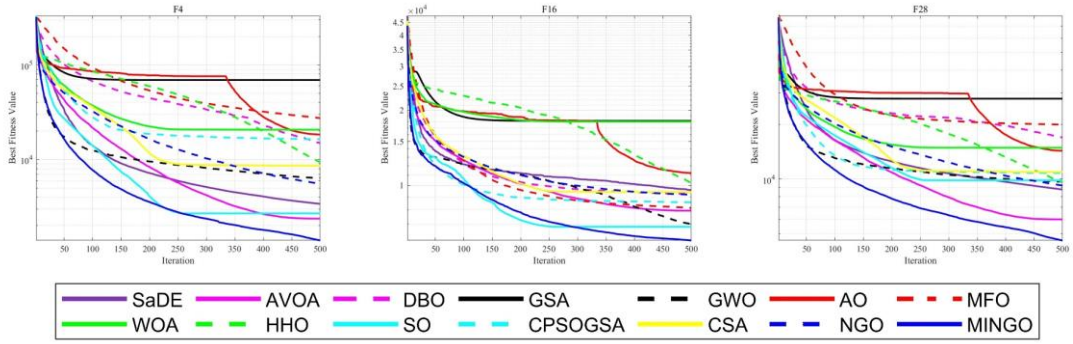

Fig. 7. CEC-2017 test function comparison curve (Dim=100).

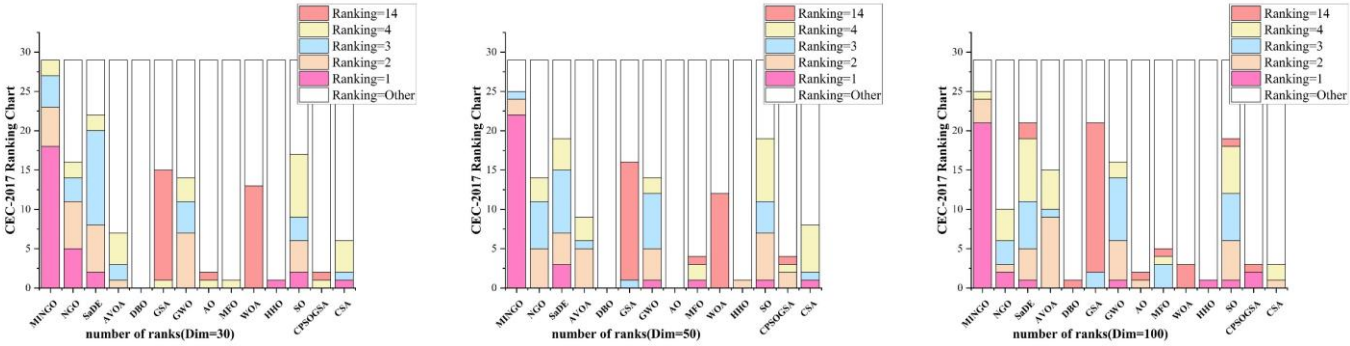

Fig. 8. Comparison of CEC-2017 test function rankings.

3.2.3. CEC-2022 Experimental results

To validate the scalability of MINGO, in this section, we will compare it with 12 other benchmark algorithms on the CEC-2022 test functions. Similar to the CEC-2017 test functions, the CEC-2022 test functions encompass unimodal functions, multimodal functions, hybrid functions, and composite functions, as detailed in Appendix 6.

The test results of MINGO and the 12 comparative algorithms on the 20-dimensional CEC-2022 test functions are presented in Appendix 7, and the convergence curves are depicted in Fig. 9. The test results demonstrate that MINGO outperforms others in optimizing 9 out of the CEC-2022 test functions in the 20-dimensional space. To provide a clearer comparison of MINGO's performance against the other algorithms in terms of rankings, we present a ranking stack plot in the form of

Fig. 10. From the graph, it can be observed that MINGO does not hold the lowest rank in any of the test functions, indicating its robust scalability and effectiveness.Considering the comprehensive rankings, MINGO attains the highest rank among the 12 algorithms, and its ranking performance significantly outperforms the other algorithms. Fig. 9illustrates the convergence curves of MINGO compared to the 12 comparative algorithms. From the graph, we can observe that MINGO exhibits higher convergence speed and accuracy compared to other algorithms. This suggests that the introduction of Brownian motion and the proximity operator in our algorithm enhances both convergence speed and accuracy.

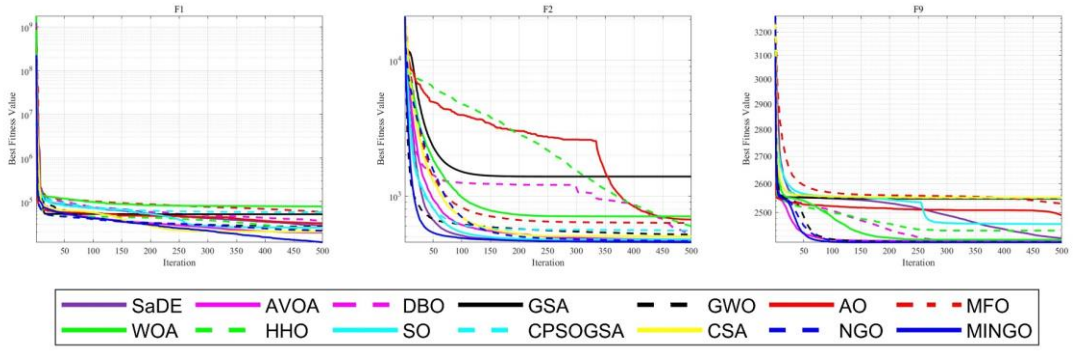

Fig. 9. CEC-2022 test function convergence curve (Dim=20).

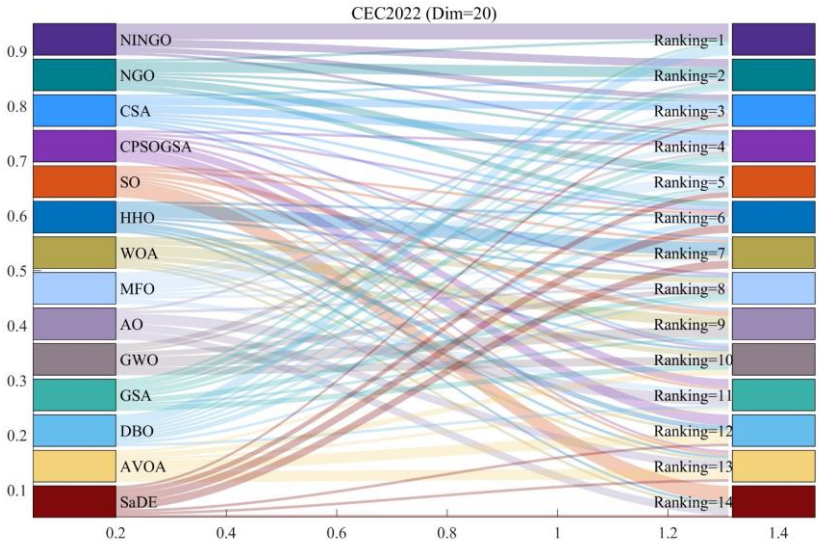

Fig. 10. Comparison of CEC-2022 test function rankings(Dim=20).

3.3. Statistical test

In this section, we will perform statistical analyses using the Wilcoxon test and the Friedman test to analyze the experimental results, assessing the differences between MINGO and the other competitive algorithms.

3.3.1. Wilcoxon test

To comprehensively demonstrate the superiority of the proposed algorithm, in this section, we will employ the Wilcoxon rank-sum test to verify whether the results of each run of MINGO differ significantly from other algorithms at a significance level of p = 5% [56]. The null hypothesis H0 is as follows: there is no significant difference between the two algorithms. When P < 5%, we reject the null hypothesis, indicating a significant difference between the two algorithms; when P > 5%, we accept the null hypothesis, suggesting that there is no significant difference between the two algorithms, i.e., the algorithm performances are comparable. N/A represents cases where the performance between the two is comparable and cannot be compared [57]. Appendix 8, Appendix 9 and Appendix 10respectively present the test results of MINGO and the comparative algorithms in CEC-2017 for dimensions 30, 50, and 100. The results for 20 dimensions in CEC-2022 are shown in Appendix 11. To highlight the comparative results, data exceeding 0.05 will be bolded in Appendix 11. To highlight the comparative results, data exceeding 0.05 will be bolded in black font.

There is no "N/A" present in either the CEC-2017 or CEC-2022 test functions, indicating that MINGO's optimization results differ significantly from those of other algorithms. From the tables, it can be observed that there is not a substantial amount of bolded data, especially in the 50 and 100-dimensional data for the 2017 test functions. This suggests that MINGO exhibits significant differences from other metaheuristic algorithms.

In conclusion, combining the analysis results from Sections 3.1 and 3.2, it is evident that MINGO demonstrates superior overall performance among various metaheuristic algorithms. This indicates the effectiveness of the four strategies we employed.

3.3.2. Friedmann test

By applying the non-parametric Friedman average rank test [58] to the experimental results of the MINGO algorithm and other algorithms on the CEC-2017 and CEC-2022 test sets, we obtained the ranking results as shown in Table. 3. Clearly, MINGO consistently ranks first, indicating that our proposed MINGO outperforms the other comparative algorithms on the considered test sets.

Table. 3. Results of the Friedman mean rank test.

Suites	CEC 2017						CEC 2022	
Dimensions	30		50		100		20	
Algorithms	Ave. Rank	Overall Rank	Ave. Rank	Overall Rank	Ave. Rank	Overall Rank	Ave. Rank	Overall Rank
SaDE	3.73	2	4.33	3	5.20	6	4.08	3
AVOA	6.83	7	5.97	6	4.40	2	5.50	5
DBO	8.90	9	9.20	9	9.17	10	7.67	8
GSA	11.83	13	11.90	13	11.43	13	11.25	12
GWO	5.60	5	5.43	5	5.07	4	6.17	6
AO	10.50	11	11.03	12	11.10	12	9.83	10
MFO	8.67	8	8.83	8	9.57	11	9.17	9
WOA	13.17	14	12.93	14	12.27	14	11.58	13
HHO	10.70	12	9.47	11	8.90	8	10.17	11
SO	4.03	4	4.10	2	4.73	3	5.00	4
CPSOGSA	9.40	10	9.23	10	9.13	9	11.58	13
CSA	6.20	6	6.30	7	6.67	7	7.58	7
NGO	3.80	3	4.57	4	5.17	5	3.58	2
MINGO	**1.63**	**1**	**1.70**	**1**	**2.20**	**1**	**1.83**	**1**

4. Application of MINGO

4.1. MINGO is used for real-world engineering optimization problems

Following the experiments and analysis in Section Four, MINGO shows superior optimization performance on test functions. To validate its practical applicability, we will apply MINGO to ten engineering problems, including three-bar truss design (TBTD), pressure vessel design (PVD), tension/compression spring design (TCPD (case1)), welded beam design (WBD), gear train design (GTD), and walking conical pulley design (SCPD)[59]. We will compare MINGO with twelve advanced algorithms: AVOA, DBO, GSA, GWO, AO, MFO, WOA, HHO, SO, CPSOGSA, CSA, and NGO. Details of the problems and test results will be provided in the following sections.

4.1.1. Three-bar Truss Design Problem (TBTD)

The three-bar truss design problem is derived from the civil engineering field, with the objective of minimizing the overall structure weight by controlling two parameter variables. The mathematical model for this problem is described by Eq. (17).

$$
\begin{aligned}
&\text{Consider} && \vec{x} = [x_1 x_2] = [A_1 A_2], \\
&\text{Minimize} && f(\vec{x}) = l * \left(2\sqrt{2x_1} + x_2\right), \\
&\text{Subject to} && g_1(\vec{x}) = \frac{\sqrt{2x_1}+x_2}{\sqrt{2x_1^2+2x_1x_2}} P - \sigma \leqslant 0, \\
& && g_2(\vec{x}) = \frac{x_2}{\sqrt{2x_1^2+2x_1x_2}} P - \sigma \leqslant 0, \\
& && g_3(\vec{x}) = \frac{1}{\sqrt{2x_2}+x_1} P - \sigma \leqslant 0, \\
&\text{Parameter range} && 0 \leqslant x_1, x_2 \leqslant 1, \\
&\text{Where} && l = 100cm, P = 2KN/cm^2, \sigma = 2KN/cm^2.
\end{aligned}
\tag{17}
$$

Table. 4 displays the optimization results of MINGO and twelve different comparative algorithms for the three-bar truss design problem. From the table, it can be observed that MINGO's results are second only to CSA, and it achieved an optimal cost of 2.6390E+02.

Table. 4. Experimental results of MINGO on a three-bar truss design.

Algorithm	Optimal values for Variable		Optimal value	Ranking
	A_1	A_2		
SaDE	7.8874E-01	4.0805E-01	2.6390E+02	3
AVOA	7.9347E-01	3.9487E-01	2.6391E+02	9
DBO	7.8854E-01	4.0863E-01	2.6390E+02	6
GSA	7.9131E-01	4.0096E-01	2.6391E+02	10
GWO	7.8994E-01	4.0471E-01	2.6390E+02	8

AO	8.0065E-01	3.8485E-01	2.6494E+02	13
MFO	7.7444E-01	4.5007E-01	2.6405E+02	12
WOA	7.7751E-01	4.4078E-01	2.6399E+02	11
HHO	7.8839E-01	4.0906E-01	2.6390E+02	4
SO	7.8831E-01	4.0928E-01	2.6390E+02	5
CPSOGSA	9.4361E-01	1.0145E-01	2.7704E+02	14
CSA	**7.8868E-01**	**4.0825E-01**	**2.6390E+02**	**1**
NGO	7.8877E-01	4.0799E-01	2.6390E+02	7
MINGO	**7.8868E-01**	**4.0825E-01**	**2.6390E+02**	**1**

The other five engineering optimization questions are listed in Appendix 12 (4.1.2-4.1.6).

In Sections 4.1.1 to 4.1.6, we conducted comparative validations of MINGO and NGO against eleven other advanced algorithms on six real engineering problems. To highlight the comparative performance, we used radar charts to visualize the rankings of each algorithm in different engineering problems, as shown in Fig. 11. A smaller area in the chart represents better performance across the six engineering problems. From the chart, it can be seen that MINGO achieved the optimal solution in every engineering problem, clearly indicating not only its outstanding performance but also high stability in solving real-world problems. The experiments in this section have thoroughly demonstrated the versatility and scalability of the MINGO method, laying a solid foundation for its practical application in engineering problem-solving.

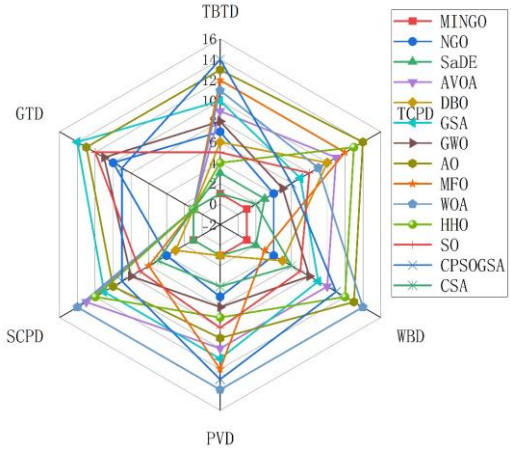

Fig. 11. Comparison of experimental results of different algorithms in 6 engineering problems.

4.2. 3D Trajectory planning for UAVs

Unmanned Aerial Vehicles (UAVs) play a vital role in various civil and military applications, and their importance and convenience are widely recognized. As a core task of the autonomous control system for UAVs, path planning and design aim to solve a complex constrained optimization problem: finding a reliable and safe path from a starting point to a goal point under certain constraints. In recent years, with the widespread application of UAVs, research on the path planning problem has garnered considerable attention. Therefore, we employ MINGO to address the UAV path planning problem and verify the algorithm's effectiveness. A specific mathematical model for this problem is outlined as follows.

4.2.1 MINGO is used to UAV 3D Path planning modeling

In mountainous environments, the flight trajectory of unmanned aerial vehicles (UAVs) is primarily influenced by factors such as high mountain peaks, adverse weather conditions, and restricted airspace areas. Typically, UAVs navigate around these regions for safety reasons during their flights. This paper focuses on the trajectory planning problem for UAVs in mountainous terrain, taking into consideration factors like high mountain peaks, weather threats, and no-fly zones, and establishes a trajectory planning model. Fig. 12illustrates the simulation environment model. The mathematical model [40, 41] for the ground and obstacle representation can be expressed by Eq. (23).

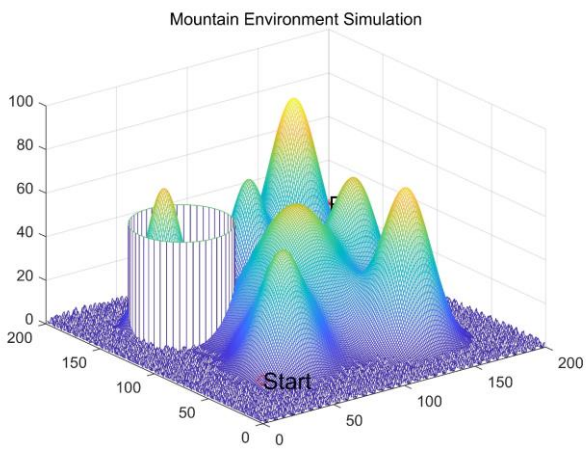

Fig. 12. Mountain Environment Simulation.

$$z = sin\ (y+1) + sin\ (x) + cos\ (x^2+y^2) + 2 \times cos\ (y) + sin\ (x^2+y^2) \tag{23}$$

During the flight of unmanned aerial vehicles (UAV), certain trajectory constraints need to be satisfied. These constraints primarily include trajectory length, maximum turning angle, and flight altitude, among others.

(1) Trajectory Length: In general, the flight of unmanned aerial vehicles (UAV) aims to minimize time and reduce costs while ensuring safety. Therefore, the path length in path planning is crucial. The mathematical model is described by Eq. (24).

$$L = \sum_{i=1}^{n-1} \parallel (x_{i+1}, y_{i+1}, z_{i+1}) - (x_i, y_i, z_i) \parallel_2 \ , (i = 2,3,\cdots,n) \tag{24}$$

The equation where (x_i, y_i, z_i) represents the ith waypoint along the planned path of the unmanned aerial vehicle.

(2) Flight Altitude: The flight altitude of the unmanned aerial vehicle significantly affects the control system and safety. The mathematical model for this constraint is shown in Eq. (25).

$$H = \sqrt{\sum_{i=1}^{n} \left(z_i - \frac{1}{n} \sum_{k=1}^{g} z_i \right)^2} \tag{25}$$

(3) Maximum Turning Angle: The turning angle of the unmanned aerial vehicle must be within a specified maximum turning angle. The constraint for the maximum turning angle can be expressed as:

$$S = \sum_{m=1}^{g} (cos\ (\varphi) - \frac{\varphi_{i+1} \times \varphi_i}{|\varphi_{i+1}| \times |\varphi_i|}) \tag{26}$$

where, φ_i is the turning angle when moving from $(x_{i+1} - x_i, y_{i+1} - y_i, z_{i+1} - z_i)$, and φ represents the maximum turning angle.

To evaluate the planned trajectory, it is common to consider multiple factors, including the maneuverability of the UAV, trajectory length, altitude above ground, and the magnitude of threats from various sources. Based on a comprehensive analysis of the impact of these factors, the trajectory cost function is calculated using the formula shown in Eq. (27) to assess the trajectory.

$$C = \sum_{i}^{n} (\omega_1 \times L + \omega_2 \times H + \omega_3 \times S) \tag{27}$$

where, L and H are the length and altitude above ground of the trajectory, respectively; S is the smoothing cost of the planned path; ω_1, ω_2 and ω_3 are weight coefficients satisfying $\omega_1 + \omega_2 + \omega_3 = 1$. By adjusting these weight coefficients, the influence of each factor on the trajectory can be modulated.

4.2.2. Example of 3D path planning for UAV

To validate the effectiveness of the proposed MINGO in three-dimensional trajectory planning for unmanned aerial vehicles (UAVs), this study conducted a verification in a simulation environment. The UAV's maximum flight altitude (H_{max}) was set to 50 m, with a flight speed of 20 m/s and a maximum turn angle (φ) of 60°. The spatial region for UAV flight in the environment was defined as a three-dimensional space with dimensions of 200 m in length, 200 m in width, and 100 m in height. The UAV's starting point was set at coordinates (0, 0, 20), and the destination point was set at (200, 200, 30). While utilizing the MINGO proposed in this study for environmental trajectory planning, a comparative analysis was performed with 12 other algorithms. The population size for all methods was set to 50, with a maximum iteration count of

500. To eliminate randomness in computational results, each algorithm was independently run 30 times under the same environmental configuration.

The statistical results are presented inAppendix 13, where "Best" represents the optimal path length, "Ave" indicates the average path length, "Worst" represents the worst path length, "Std" denotes the standard deviation, "Median" is the median, and "Rank" signifies the ranking based on the average path length. Fig. 13illustrates the search iteration curve for obtaining the optimal fitness value, while Fig. 14provides 3D and 2D schematic diagrams for the optimal range obtained by the 13 different methods.

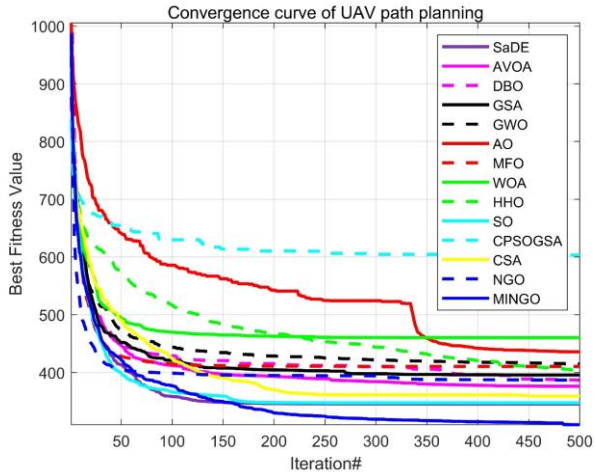

Fig. 13. Optimal fitness search iteration curve.

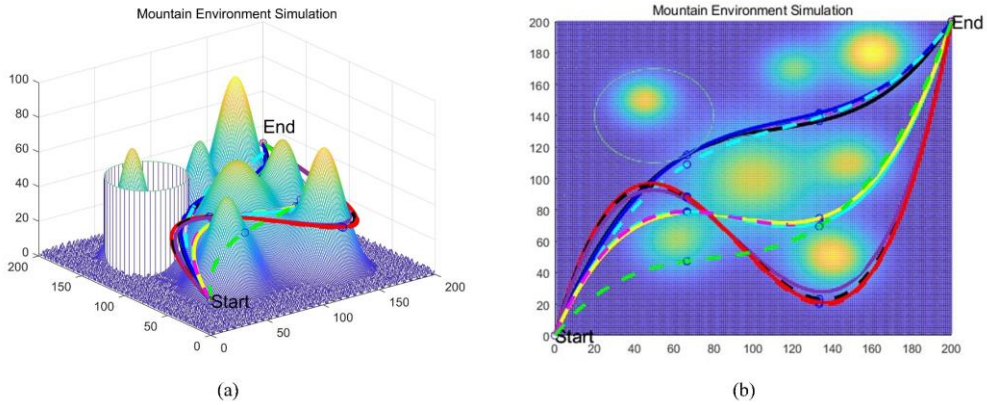

Fig. 14. Flight tracks optimized by algorithms.

From Appendix 3, it can be observed that, in terms of both the optimal fitness value and the average fitness value, the MINGO method proposed in this paper yields the smallest fitness value, while CPSOGSA produces the largest. Additionally, the average fitness value of MINGO is even smaller than the optimal fitness values of the other 12 methods, indicating higher computational precision for MINGO. In terms of standard deviation, MINGO has the smallest value, suggesting stronger computational stability compared to the other 12 methods.

The search iteration curve inFig. 13. Specifically, compared to methods such as NGO and others, MINGO exhibits a faster convergence rate and higher convergence accuracy. As evident in Fig. 14-(a), the trajectories planned by the 13 methods effectively avoid threat areas, demonstrating the feasibility of the generated trajectories. In Fig. 14-(b),the trajectory obtained by MINGO is the shortest and relatively smoother, while CPSOGSA's trajectory is the worst, requiring ascent to a certain height and navigating a certain distance to avoid threat areas. The results indicate that MINGO can effectively enhance the efficiency of trajectory planning, demonstrating a certain advantage.

5. Conclusion and Outlook

In response to the deficiencies in the Northern Goshawk Optimization (NGO) algorithm, such as an imbalance between exploration and exploitation, slow convergence speed, susceptibility to local optima, and low convergence accuracy, this paper proposes an enhanced version of NGO through the fusion of multiple strategies. The improvements primarily focus on the exploration strategy and the mode of exploitation. By enhancing the search formula during the exploration phase, a better balance between exploration and exploitation is achieved. Building upon this, the Levy flight strategy is introduced to maintain population diversity during the exploitation phase, thereby enhancing convergence accuracy. Furthermore, to prevent the search process from becoming trapped in local optima, the Cauchy mutation strategy and Brownian motion strategy are also incorporated to improve the algorithm's ability to escape local optima during the problem search process.

To validate the rationale behind the improvement strategies and the effectiveness of the proposed algorithm, an

analysis was conducted concerning exploration and exploitation, population diversity, and a comparative study with different strategy algorithms. The analysis results indicate that MINGO excels in maintaining a balance between exploration and exploitation, as well as preserving population diversity. Furthermore, the introduction of various strategies demonstrates the algorithm's capability to enhance specific aspects where NGO was lacking.

To assess the performance of MINGO, experiments were conducted using the CEC2017 and CEC-2022 benchmark test functions. The results indicate that MINGO exhibits a clear advantage, whether in different dimensions of CEC-2017 or on the CEC-2022 test functions. Importantly, MINGO maintains strong problem-solving capabilities even in high-dimensional problems. Furthermore, to validate the algorithm's ability to address real-world problems, it was applied to six classic engineering problems and a 3D Trajectory planning for UAVs.The test results demonstrate that MINGO maintains high optimization capabilities in solving constrained problems, and compared to other algorithms, yields the best optimization results for these problems. In summary, MINGO, proposed in this paper as an enhancement of the Northern Goshawk Optimizer, has shown significant performance improvements. It performs well in both unconstrained and constrained problems, showcasing its robustness and wide applicability.

Although our method achieves satisfactory optimization results in most CEC2017, CEC2022, engineering design optimization problems, and UAV path planning, it must be acknowledged that certain limitations exist. Particularly, in complex functions such as F3, F10, and F18 in CEC2017, our method still tends to get stuck in local optima. This challenge is not exclusive to CEC competitions but can also arise in other scenarios like large-scale optimization problems and job shop scheduling.

To address these challenges, continuous improvement of our method is necessary to adapt to diverse optimization problems effectively. Firstly, we can introduce more diversity maintenance mechanisms to prevent falling into local optima. This could involve employing multiple heuristic search strategies or enhancing population diversity. Secondly, we could explore more sophisticated optimization algorithms, such as those based on swarm intelligence, deep learning, or genetic algorithms, to enhance global search capabilities. Additionally, for different problem types, designing specific problem structures and constraints could better guide the optimization process.

In conclusion, for optimization tasks prone to local optima issues, ongoing refinement of our method and tailored adjustments based on problem characteristics are essential to enhance optimization efficacy and tackle complex problems effectively.

In the future, we intend to apply MINGO to address various practical problems, including but not limited to workshop scheduling, image processing, robot path planning, and tool fault diagnosis.

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Data Availability Statement

All data generated or analysis during this study are included in this published article.

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